

NOTE: All items identified by description. Part numbers to be assigned at purchase-order stage. ★ = Long-lead / critical-path item — order first.

COMMERCIAL BILL OF MATERIALS

#	Description	Qty	Unit	Technical Specification	Delivery priority	Notes
SH AF T						
1	★ Head shaft (solid section)	1	EA	Ø108 × 858 mm lg	★ CRITICAL	Governed by stress; deflection — mm
2	★ Boot shaft (solid section, free-running)	1	EA	Ø100 × 858 mm lg	Long lead	Governed by stress — no drive torque, no keyway required
3	★ Head shaft key	1	EA	25×14mm × 136mm lg	★ CRITICAL	PASS — shear 18.6 MPa, bearing 66.4 MPa within limits.
PU LL EY						
4	Head pulley shell	1	EA	Ø500.0 × 406 mm F2F, t=11mm	Standard	
5	Head pulley end disc (annular plate)	2	EA	OD=478mm ID=108mm t=26mm	Standard	Specify 26mm (+20% over 22.0mm calc minimum)
6	Head pulley lagging — ceramic embedded rubber	1	EA	t=16mm OD=532mm	Standard	μ operating = 0.5
7	Boot / tail pulley assembly (shell + discs)	1	EA	Ø300 × 406 mm F2F	Standard	Take-up point; snub or gravity arrangement
BE LT						
8	★ Elevator belt — 2 PLY	1	SET	BW=356mm L=53m mass≈422kg	★ CRITICAL	Include 2 splices; splice type per belt manufacturer recommendation
9	Buckets — Series AC_12x8 305×216mm	225	EA	305×216×203mm 8.58L	Standard	Spacing = 228mm c/c ×225 off
10	Bucket attachment bolt set — M12 stainless	225	SET	M12 × 35 HHB + washer + nylock nut (2 per bucket)	Standard	Replace at every bucket change; inspect at 8,000h
DR IV E						
11	★ Drive motor — 37 kW IE3	1	EA	37kW 280/140 rpm B3/B5 frame	★ CRITICAL	SF = 1.25; confirm frame and mounting with supplier
12	★ Helical-bevel gearbox	1	EA	i = 20.7:1 T _{out} = 3413.9 Nm 70 rpm output	★ CRITICAL	Confirm service factor with gearbox manufacturer; oil bath lubrication
TA KE -U P						

#	Description	Qty	Unit	Technical Specification	Delivery priority	Notes
13	Gravity take-up counterweight (ballast + frame)	1	SET	Total mass = 3597 kg (frame 1079 kg + ballast 2518 kg)	Standard	Travel = 461 mm min; allow 20% headroom for field adjustment
CASING						
14	Casing plate (fabricated panels)	1	SET	t = 8mm area ≈ 34 m ²	Standard	Stiffener pitch = 1755mm; min 14 stiffener sets over height
FASTENERS						
15	Casing assembly bolt set — M12	793	SET	M12 hex bolt + nut + washer, panel-to-panel and panel-to-stiffener fixing	Standard	667 seam + 126 stiffener-band bolts; first-order fabrication estimate, not a structural connection design — confirm against actual panel module size
BEARINGS						
16	★ Head shaft pillow block bearing assembly	2	EA	Shaft Ø108mm L10 = 180,226h at 70rpm	★ CRITICAL	SNL plummer block with labyrinth seal; grease-lubricated
17	★ Boot shaft pillow block bearing assembly	2	EA	Shaft Ø100mm L10 = 322,687h at 70rpm	Long lead	SNL plummer block with labyrinth seal; grease-lubricated
CHUTE						
18	Discharge chute fabrication (back, side plates)	1	SET	W=605mm angle=60.2°	Standard	
19	Chute wear liner — AR500 Wear Plate	1	SET	t=12mm W=605mm	Standard	Wear index = 16.79 (HIGH); plugging risk = HIGH

LONG-LEAD / CRITICAL-PATH SUMMARY

Item	Typical lead time	Action required
Motor	8–16 weeks (IEC high-efficiency)	Confirm frame, voltage, IE class with supplier
Gearbox	10–20 weeks (configured unit)	Confirm ratio, mounting, shaft dimensions
VFD / Drive	6–14 weeks	Confirm kW, voltage, comms protocol
Elevator belt	4–8 weeks (cut-to-length)	Confirm BW, ply, cover grade, splice type
Chain (ER/SC)	6–10 weeks (import)	Confirm pitch, WL, coating
Head shaft	4–6 weeks (machined)	Issue machining drawing with MTC requirement
Head bearing	2–4 weeks	Confirm bore, PN, quantity

PACKAGING & TRANSPORT NOTES

- Casing sections: split into max 6 m lengths for transport. Approx 4 lifts.
- Head section (approx 756 mm wide): transport as single assembled unit if road permits.

- Belt: ship on drum, diameter per supplier minimum bend radius. Do NOT roll flat.
- Buckets: pack in bulk crates, max stack height 1.2 m, no sharp edges exposed.
- Shaft: ship in timber cradle, protected from impact. Mark 'DO NOT STACK'.
- All items to be tagged with model number and item description matching this BOM.